FREDY LUNG

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EDUCATION

California State University, Los Angeles | May 2027 | B.S. in Computer Science | GPA: 3.93

Projects

SQL database |*C++* | *Data Structures* | *File I/O* | *B+ Trees* | *FSM Parsing*

- Built a mini-SQL database engine supporting SELECT, INSERT, and CREATE TABLE queries via a custom lexer, parser, and execution engine
- implemented a persistent B+ tree index for search and range queries; parsed conditional expressions using finite-state tokenization and Shunting Yard evaluation
- Developed custom Map and MultiMap containers to manage indexed records and query processing

Book Library Web App | *Django* | *Python* | *Bootstrap* |*SQLite*

- Developed a full-stack book platform with authentication, search, categories, ratings, favorites, and a session-based shopping cart
- Used Django ORM, ModelForms, and reusable templates to support database operations, form validation, and responsive interfaces
- Added book commenting, user-managed listings, category filters, and authenticated favorites to support core library workflows
- Built responsive UI with Bootstrap supporting mobile and desktop layouts

SMPTE Media Standards Senior Design Project | 2026-2027 | In Progress

- Collaborate on an industry-sponsored senior design project to scope a standards-based media measurement workflow and define engineering requirements
- Research device capabilities, software interfaces and SDK availability to assess integration options and project feasibility

WORK EXPERIENCE

ASCE Pacific Southwest Symposium 2026 at CalStateLA– Web Designer **September 2024 – April 2026**

- Maintained and optimized event website performance, ensuring fast load times and responsive design
- Implemented front-end updates using HTML, CSS, and JavaScript to improve user experience
- Collaborated with cross-functional teams to translate requirements into functional web features

California State University, Los Angeles
Mathematics Instructional Student Assistant. **June 2025 - Present**

- Co-lead and mentor groups of 30 students in a Pre-Calculus, Calculus I, and Calculus II workshop
- Collaborate with faculty and students to implement strategies to improve student engagement and performance
- Evaluate student assignments and present weekly reports about students' performance

LEADERSHIP

American Society of Civil Engineers (ASCE), Cal State LA — Webmaster **May 2026 - Present**

- Build and maintain the ASCE Cal State LA chapter website using WordPress, updating page layouts, navigation, and published content
- Developed and organized Home, Projects, About, Resources, Membership, Officers, Contact, and Sponsors pages to improve information access
- Customize site navigation, dropdown menus, header/footer elements, and CSS styling; coordinate content updates with chapter leadership

American Society of Civil Engineers (ASCE), Cal State LA — Webmaster **May 2026 - Present**

- Manage and update the chapter website to support events, announcements, and member resources
- Collaborate with the executive board to maintain the chapter's digital presence and communicate organizational initiatives

Technical Interview Preparation for Software Engineers (TIPSE) — Sponsor Outreach **May 2026 - Present**

- Conduct outreach to companies and industry professionals to develop sponsorship opportunities and external partnerships
- Coordinate with the organization's leadership team to support technical interview preparation events and student programming

Skills

Languages: C/C++, Java, Python, SQL, HTML/CSS, C#, Kotlin
Frameworks: Spring Boot, Django, Bootstrap
Tools: Git, GitHub, Docker, Jira, Linux, LaTex, Claude Code, Open AI Codex, Office 365, Canva, Unity, WordPress
Databases & Concepts: MySQL, SQLite, B+ Trees, Data Structures, Algorithms, REST APIs, MVC, File I/O, session management
Languages (Spoken): Spanish, Mandarin, Cantonese